An Interpretable Machine Learning Model for Predicting Muscle Loss Risk in Oral Cancer Patients Post-Radiotherapy Using Pre-treatment Clinical and Dosimetric Features

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Introduction

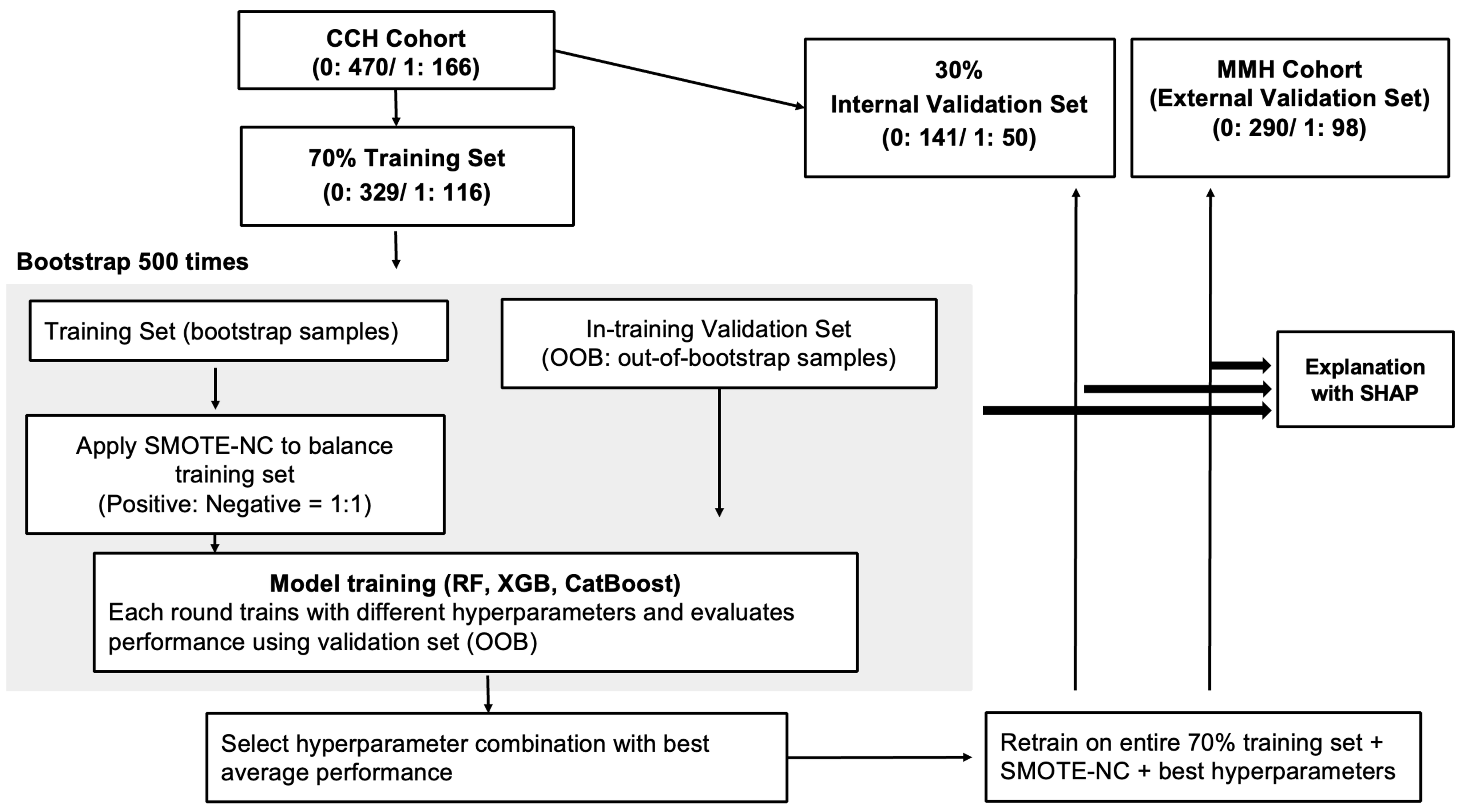
- Oral cavity cancer remains a critical global public health concern. According to the 2023 Taiwan Cancer Registry report, it ranks fourth in both incidence and mortality among the top ten cancers in males.
- Although radiotherapy effectively controls tumor progression, it causes healthy tissue damage, leading to side effects like oral mucositis, dysphagia, and fatigue.
- The DARS phase 3 trial confirms a dose-dependent link between radiation to the pharyngeal constrictor muscles (PCM) and severe dysphagia[1], which frequently triggers a clinical cascade toward malnutrition and systemic muscle loss.
- Muscle loss has emerged as a paramount clinical concern, as it is closely associated with reduced treatment tolerance, increased postoperative complications, and diminished overall survival.
- **Aim:** To develop a pre-treatment machine learning model integrating clinical and dosimetric features for the early identification of patients at high risk for muscle loss.

Methods

Population: This retrospective multicenter study enrolled 1,024 oral cancer patients (2010–2021) from Changhua Christian Hospital (CCH, n = 636) and MacKay Memorial Hospital (MMH, n = 388), after excluding patients with prior malignancies or incomplete clinical and follow-up data.

Muscle Loss Definition: Significant muscle loss was defined as a ≥ 4.2% decrease in L3-equivalent Skeletal Muscle Index (SMI) converted from C3-level CT images, a threshold associated with significantly increased all-cause mortality[2].

Model Features: Models integrated clinical variables (age, sex, ECOG performance status, tumor site, pathological stage, and chemotherapy) and prognostic indicators, including Neutrophil-to-Lymphocyte Ratio (NLR), Charlson Comorbidity Index (CCI), Mini Nutritional Assessment (MNA), and pre-treatment SMI. Furthermore, dosimetric factors of swallow-related structures were incorporated, specifically the radiation doses to the Cervical Esophagus (CE), Cricopharyngeus Muscle (CPA), Esophagus Inlet Muscles (EIM), Glottic Larynx (GL), Supraglottic Larynx (SGL), and the Superior, Middle, and Inferior Pharyngeal Constrictor Muscles (SPCM, MPCM, IPCM).



Results

Training Set

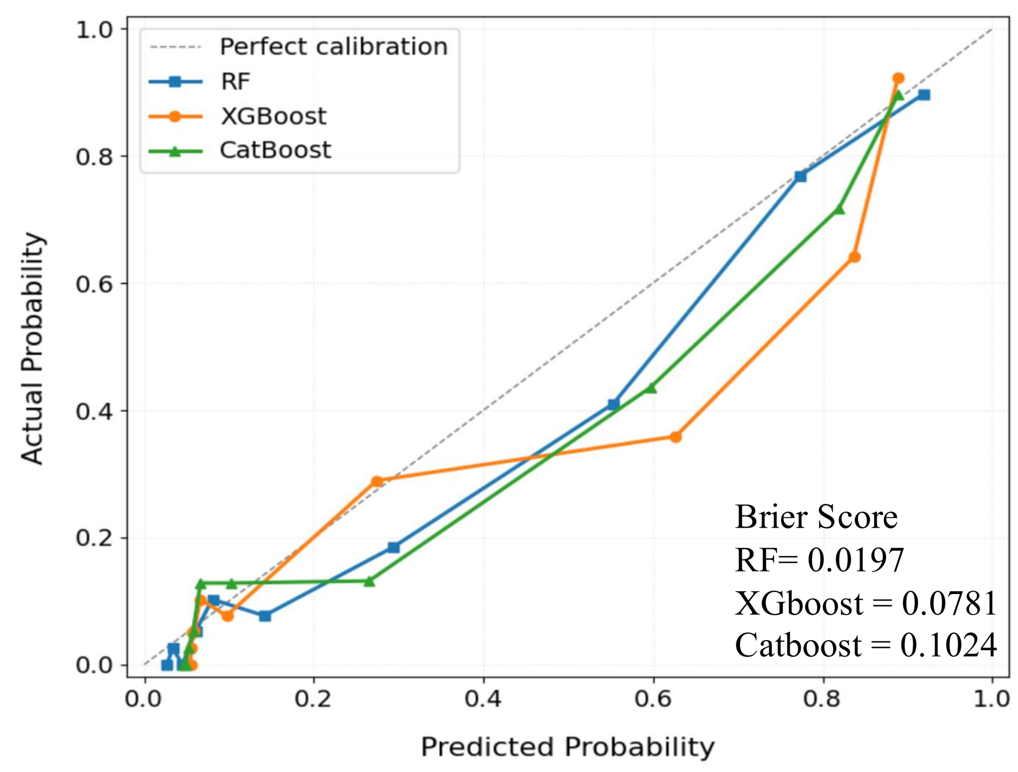
Model	F1 score	ROC-AUC	PR-AUC	Sensitivity	Specificity	Precision	Accuracy
	(95% CI)	(95% CI)	(95% CI)	(95% CI)	(95% CI)	(95% CI)	(95% CI)
RF	0.776^{b}	0.935	0.862^{b}	0.769^{b}	0.927^{a}	0.789^{a}	0.885^{c}
	(0.702, 0.840)	(0.903, 0.963)	(0.798, 0.916)	(0.639, 0.880)	(0.879, 0.969)	(0.691, 0.889)	(0.849, 0.920)
XGBoost	0.775^{b}	0.934	0.867^{a}	0.793^{a}	0.911^{c}	0.762^{b}	0.880^{b}
	(0.707, 0.837)	(0.903, 0.961)	(0.810, 0.921)	(0.688, 0.893)	(0.857, 0.957)	(0.657, 0.868)	(0.845, 0.914)
CatBoost	0.786^{a}	0.936	0.865^{a}	0.790^{a}	0.923^{b}	0.786^{a}	0.888^{a}
	(0.720, 0.848)	(0.904, 0.963)	(0.802, 0.913)	(0.678, 0.888)	(0.873, 0.964)	(0.690, 0.886)	(0.855, 0.922)

Differences between models within the same column were assessed using Kruskal-Wallis and Dunn's post-hoc tests. Values with different superscript letters are significantly different ($p < 0.05$).

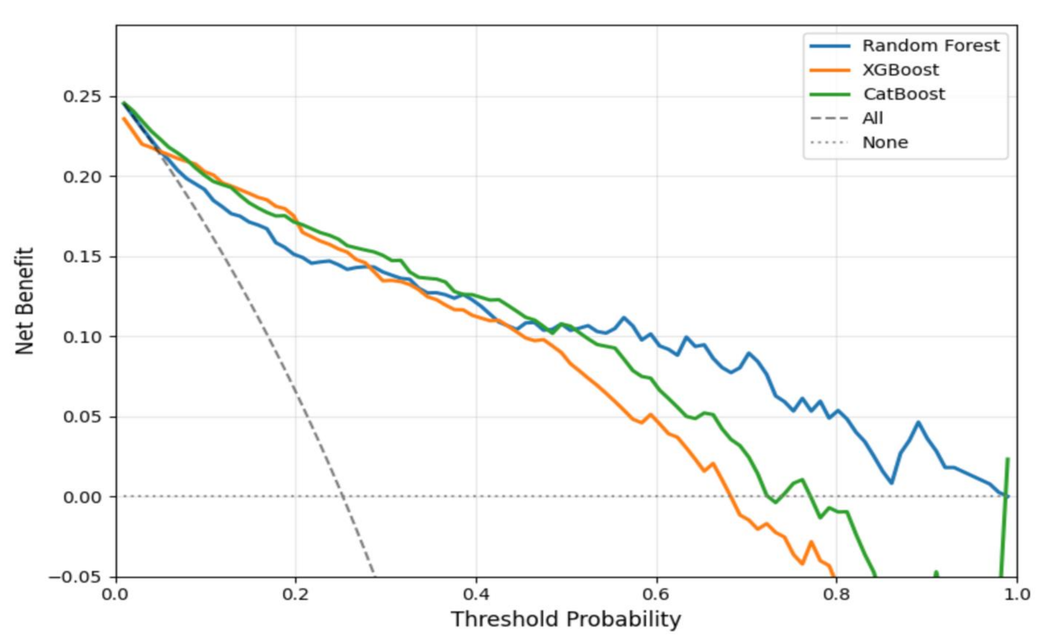
Internal & External Validation Set

Model	F1 score	ROC-AUC	PR-AUC	Sensitivity	Specificity	Precision	Accuracy
Internal Validation							
RF	0.800	0.960	0.850	0.880	0.887	0.733	0.885
XGBoost	0.785	0.948	0.812	0.840	0.894	0.736	0.880
CatBoost	0.793	0.954	0.843	0.880	0.879	0.721	0.880
External Validation							
RF	0.740	0.913 *	0.813	0.827	0.862	0.669	0.853
XGBoost	0.704	0.892	0.789	0.776	0.855	0.644	0.835
CatBoost	0.738	0.904	0.800	0.816	0.866	0.672	0.853

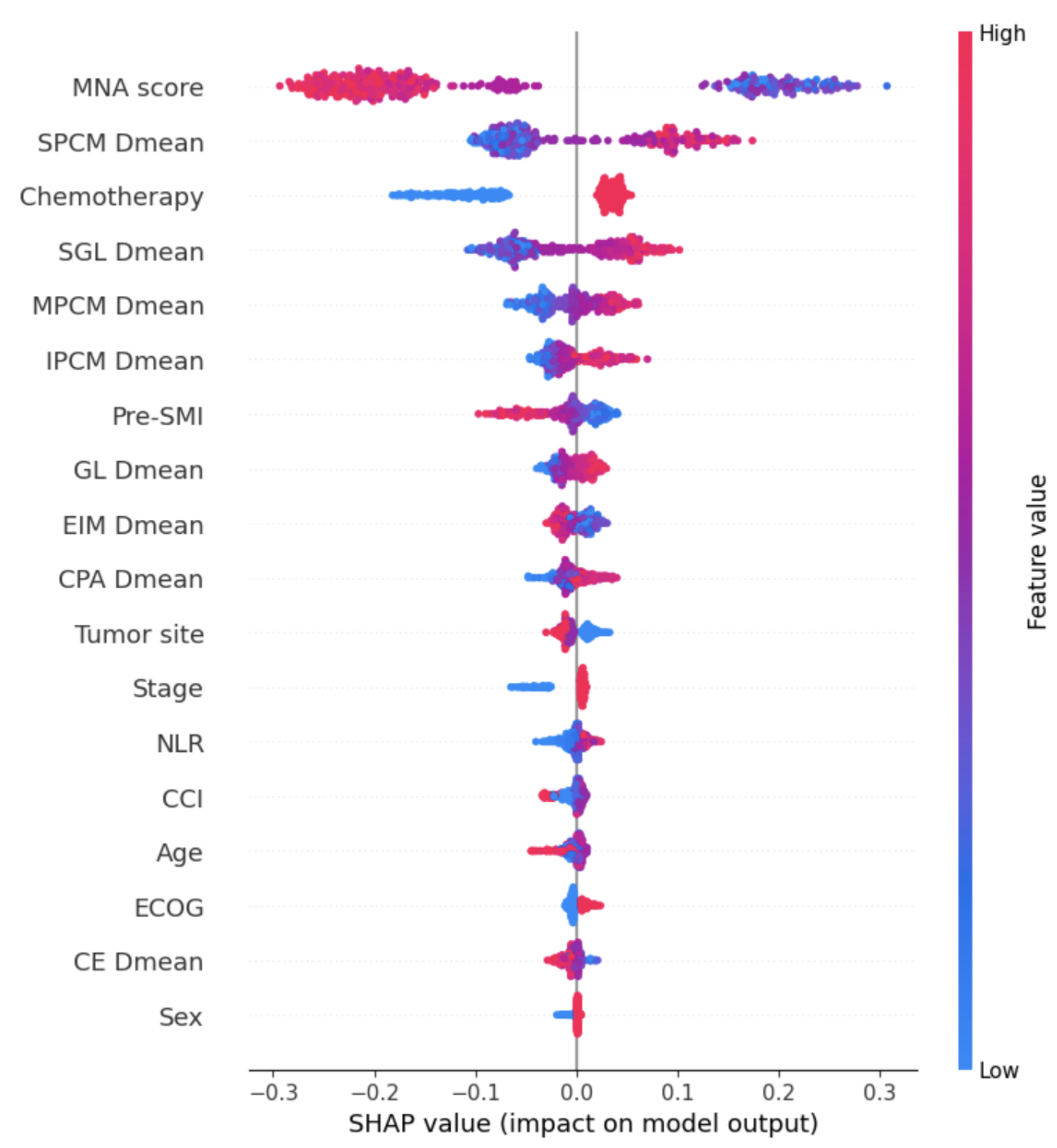
* Significant difference between RF and XGBoost (DeLong's test).



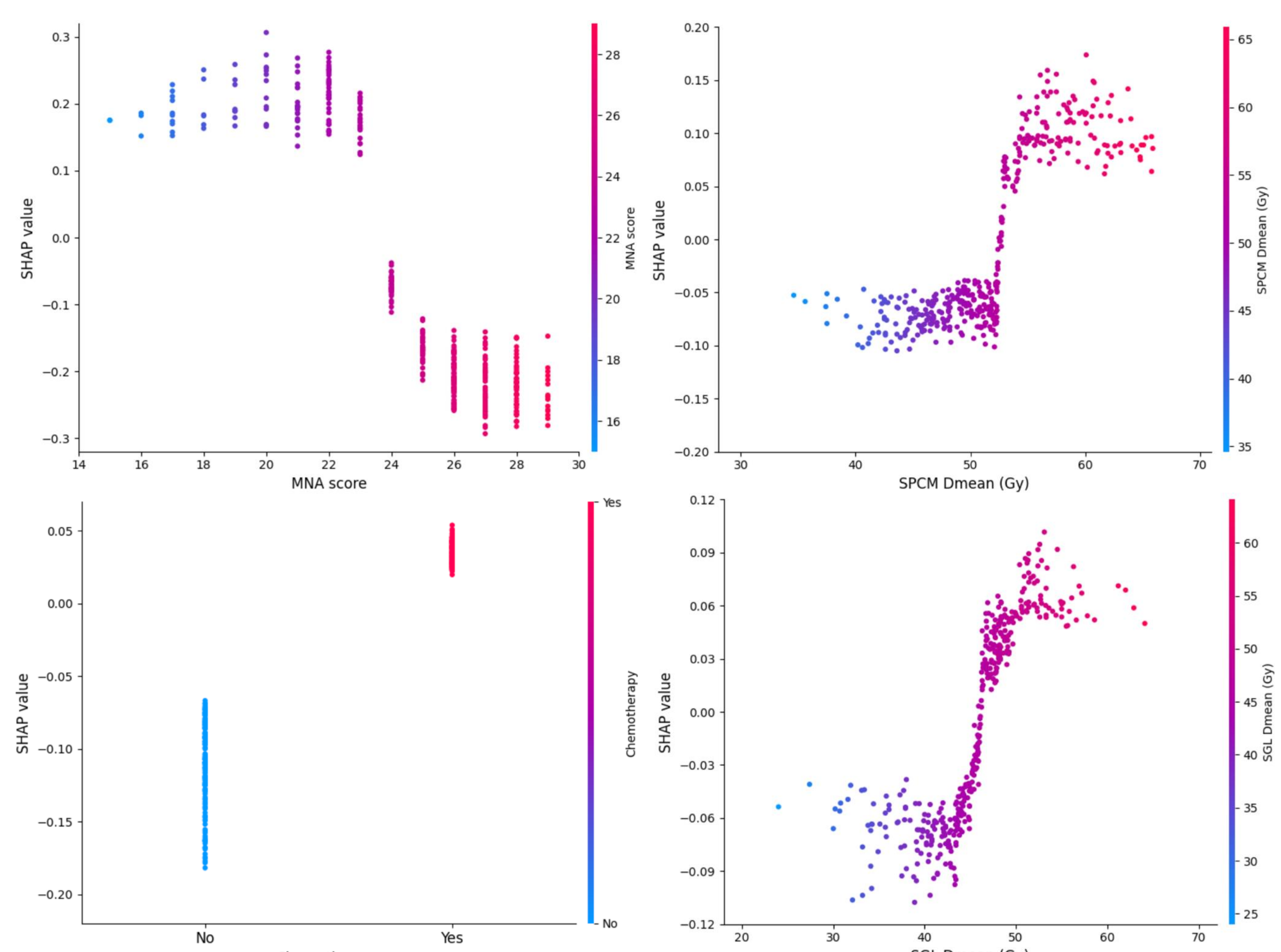
Calibration Curve



Decision Curve Analysis



SHAP Summary Plot of RF



SHAP Dependence Plot of RF

Conclusions

- **RF Model Superiority**: The Random Forest model outperformed others with an external validation AUC of 0.913, demonstrating superior calibration and clinical net benefit (DCA).
- **Key Predictors**: SHAP analysis identified MNA score as the primary predictor, with radiotherapy doses to swallowing structures (SPCM, SGL) and chemotherapy as critical risk drivers for muscle loss.
- **Future Research**: Prospective, multicenter trials across diverse head and neck cancer subtypes are needed to validate the model's generalizability and clinical utility for early intervention.

References:
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2. Lee, J., Lin, J.-B., Lin, W.-C., Jan, Y.-T., Leu, Y.-S., Chen, Y.-J., & Wu, K.-P. (2025) Identifying threshold of CT-defined muscle loss after radiotherapy for survival in oral cavity cancer using machine learning. *European Radiology*, *35*(7), 4289-4299. https://doi.org/10.1007/s00330-024-11303-4